Optimization Methods in Machine Learning, CUB, Spring 2026

Theory 1. Convergence speed and second differential.

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1. Problem 1

For each of the following sequences $\{r_k\}$ find out its convergence speed (linear, sublinear, superlinear). In case of superlinear rate find out additionally whether the sequence has quadratic convergence rate.

1.1. (a) $r_k = (0.99)^{k^2}$

$$\frac{r_{k+1}}{r_k} = \frac{(0.99)^{(k+1)^2}}{(0.99)^{k^2}} = (0.99)^{2k+1}.$$

Since $\lim_{k\to\infty} (0.99)^{2k+1} = 0$, the convergence is superlinear.

$$\frac{r_{k+1}}{r_k^2} = (0.99)^{(k+1)^2-2k^2} = (0.99)^{-k^2+2k+1} \to \infty,$$

since the exponent $-k^2 + 2k + 1 \to -\infty$ and the base $0.99 < 1$.

Superlinear convergence (not quadratic).

1.2. (b) $r_k = (0.99)^{2^k}$

$$\frac{r_{k+1}}{r_k} = \frac{(0.99)^{2^{k+1}}}{(0.99)^{2^k}} = (0.99)^{2^k} \to 0.$$

The convergence is superlinear. Check quadratic:

$$\frac{r_{k+1}}{r_k^2} = \frac{(0.99)^{2^{k+1}}}{\left((0.99)^{2^k}\right)^2} = \frac{(0.99)^{2^{k+1}}}{(0.99)^{2^{k+1}}} = 1.$$

The limit is exactly 1 (a positive constant).

Quadratic convergence.

1.3. (c) $r_k = \frac{1}{k!}$

$$\frac{r_{k+1}}{r_k} = \frac{k!}{(k+1)!} = \frac{1}{k+1} \to 0.$$

Superlinear. Check quadratic:

$$\frac{r_{k+1}}{r_k^2} = \frac{(k!)^2}{(k+1)!} = \frac{k!}{k+1} \to \infty.$$

Superlinear convergence (not quadratic).

1.4. (d) $r_k = \frac{1}{k^k} = k^{-k}$

$$\frac{r_{k+1}}{r_k} = \frac{k^k}{(k+1)^{k+1}} = \frac{1}{k+1} \cdot \frac{1}{(1+1/k)^k} \to 0 \cdot \frac{1}{e} = 0.$$

Superlinear. Check quadratic:

$$\ln \frac{r_{k+1}}{r_k^2} = 2k \ln k - (k+1)\ln(k+1) \approx k \ln k \to \infty.$$

Superlinear convergence (not quadratic).

1.5. (e)

$$r_k = \begin{cases} (0.99)^{2^k} & \text{if } k \text{ is even} \\ r_{k-1}/k & \text{if } k \text{ is odd.} \end{cases}$$

-k even:

$$\frac{r_{k+1}}{r_k} = \frac{r_k/(k+1)}{r_k} = \frac{1}{k+1} \to 0.$$

-k odd, so $k+1$ even:

$$\frac{r_{k+1}}{r_k} = \frac{(0.99)^{2^{k+1}}}{(0.99)^{2^{k-1}}/k} = k \cdot (0.99)^{2^{k+1}-2^{k-1}} \to 0.$$

So the convergence is superlinear. Check quadratic at k even:

$$\frac{r_{k+1}}{r_k^2} = \frac{r_k/(k+1)}{r_k^2} = \frac{1}{(k+1)r_k} = \frac{1}{(k+1)(0.99)^{2^k}} \to \infty.$$

Superlinear convergence (not quadratic). The even subsequence is quadratic, but the odd steps break the quadratic rate for the full sequence.

2. Problem 2

Suppose that $\{r_k\}$ is a sequence of non-negative values given by $r_{k+1} = Mr_k^2$, where $M > 0$, $r_0 \geq 0$. Find necessary and sufficient condition for M and r_0, when the sequence converges to zero. What would be the convergence rate in this case?

Take logarithms: let $y_k = \ln r_k$:

$$y_{k+1} = 2y_k + \ln M.$$

The general solution is $y_k = c \cdot 2^k - \ln M$ with $c = \ln(Mr_0)$. Exponentiating:

$$r_k = \frac{1}{M}(Mr_0)^{2^k}.$$

$r_k \to 0$ if and only if $|Mr_0| < 1$, i.e.:

$$r_0 < \frac{1}{M}.$$

$$\frac{r_{k+1}}{r_k^2} = \frac{Mr_k^2}{r_k^2} = M = \text{const} > 0.$$

Quadratic convergence when $r_0 < 1/M$.

3. Problem 3

For each of the following functions find out the second differential $d^2 f(x)[dx, dx]$ and show that this differential has constant sign. Find out also whether this sign is strict for all $dx \neq 0$ or not.

3.1. (a) $f(x) = \ln\left(-\frac{1}{2}x^\top Ax - b^\top x - c\right)$, where A is non-negatively definite and the function is considered only for those values of x where the expression under logarithm is strictly positive.

Let $u(x) = -1/2x^\top Ax - b^\top x - c > 0$.

$$df = \frac{du}{u} = \frac{-(x^\top A + b^\top)dx}{u}.$$

$$d^2 f = \frac{u \cdot (-dx^\top Adx) - \left(-(Ax+b)^\top dx\right)^2}{u^2}.$$

$$d^2 f[dx, dx] = -\frac{1}{u^2}\left[u(dx^\top Adx) + \left((Ax+b)^\top dx\right)^2\right].$$

Since $A \succeq 0$ we have $dx^\top Adx \geq 0$, and $u > 0$, and the squared term ≥ 0. The overall factor is $-1/u^2 < 0$.

$$d^2 f[dx, dx] \leq 0 \quad \Longrightarrow \quad \text{concave}.$$

The sign is not necessarily strict: if $dx \in \ker A$ and $(Ax+b)^\top dx = 0$, then $d^2 f = 0$ for $dx \neq 0$.

3.2. (b) $f(x) = \left(\sum_{i=1}^n x_i^p\right)^{1/p}$, where $p < 1$, $p \neq 0$, $x_i > 0$ for all i.

Let $S = \sum x_i^p$, so $f = S^{1/p}$.

$$df = S^{1/p-1} \sum x_i^{p-1} dx_i.$$

$$d^2 f = (1-p)S^{1/p-2}\left[\left(\sum x_i^{p-1} dx_i\right)^2 - S \sum x_i^{p-2}(dx_i)^2\right].$$

By Cauchy–Schwarz with $u_i = x_i^{p/2}$, $v_i = x_i^{p/2-1} dx_i$:

$$\left(\sum u_i v_i\right)^2 \leq \left(\sum u_i^2\right)\left(\sum v_i^2\right) \quad \Longrightarrow \quad \left(\sum x_i^{p-1} dx_i\right)^2 \leq S \cdot \sum x_i^{p-2}(dx_i)^2.$$

The bracket is ≤ 0, the prefactor $(1-p)S^{1/p-2} > 0$ (since $p < 1$).

$$d^2 f[dx, dx] \leq 0 \quad \Longrightarrow \quad \text{concave}.$$

The sign is strict for $dx \neq 0$ unless $dx_i \propto x_i$ (Cauchy–Schwarz equality condition).

3.3. (c) $f(X) = \operatorname{tr}(X^{-1}A)$, where X is positive definite and A is non-negative definite.

Using $d(X^{-1}) = -X^{-1}(dX)X^{-1}$:

$$df = -\operatorname{tr}(X^{-1}AX^{-1}dX).$$

$$d^2 f = 2\operatorname{tr}(X^{-1}dXX^{-1}AX^{-1}dX).$$

Let $Y = X^{-1/2} dX X^{-1/2}$ and $B = X^{-1/2} A X^{-1/2} \succeq 0$. Then:

$$d^2 f = 2\,\mathrm{tr}(BY^2) \geq 0,$$

since $B \succeq 0$ and $Y^2 \succeq 0$, and the trace of two PSD matrices is non-negative.

$$d^2 f[dX, dX] \geq 0 \quad \Longrightarrow \quad \text{convex}.$$

If $A \succ 0$ then $B \succ 0$, and $\mathrm{tr}(BY^2) = 0$ only when $Y = 0$ (i.e. $dX = 0$), so the sign is strict. If A is only $\succeq 0$ (singular), the sign is not necessarily strict.

3.4. (d) $f(X) = (\det X)^{1/n}$, where X is positive definite.

Using $d(\det X) = (\det X)\,\mathrm{tr}(X^{-1} dX)$:

$$df = \frac{1}{n} f(X)\,\mathrm{tr}(X^{-1} dX).$$

$$d^2 f = \frac{f(X)}{n} \left[\frac{1}{n} \left(\mathrm{tr}(X^{-1} dX) \right)^2 - \mathrm{tr}\left((X^{-1} dX)^2 \right) \right].$$

Let $M = X^{-1/2} dX X^{-1/2}$ with eigenvalues $\lambda_1, ..., \lambda_n$. Then $\mathrm{tr}(M) = \sum \lambda_i$ and $\mathrm{tr}(M^2) = \sum \lambda_i^2$. By Cauchy–Schwarz:

$$\frac{1}{n} \left(\sum \lambda_i \right)^2 \leq \sum \lambda_i^2.$$

So the bracket is ≤ 0, and $f(X)/n > 0$.

$$d^2 f[dX, dX] \leq 0 \quad \Longrightarrow \quad \text{concave}.$$

The sign is strict for $dX \neq 0$ unless all λ_i are equal, i.e. $dX \propto X$.

4. Problem 4

Let $f(x) = \frac{x^\top A x}{\|x\|^2}$, where A is a symmetric matrix and $x \neq 0$. Show that all stationary points of this function are determined by eigenvectors of the matrix A. Find the Hessian $\nabla^2 f(x)$ for eigenvectors x. Determine the type of stationarity (local minima, local maxima, saddle point).

$$\nabla f(x) = \frac{2}{x^\top x} (Ax - f(x) x).$$

Setting $\nabla f = 0$ gives $Ax = \lambda x, \quad \lambda = f(x)$. The stationary points are the eigenvectors of A, and the values are the corresponding eigenvalues.

At a stationary point v with $Av = \lambda v$, $\nabla f(v) = 0$:

$$\nabla^2 f(v) = \frac{2}{v^\top v} (A - \lambda I).$$

Let $\lambda_1 \leq \lambda_2 \leq ... \leq \lambda_n$ be the eigenvalues of A.

-$\lambda = \lambda_1$ (minimum eigenvalue): $A - \lambda_1 I$ has eigenvalues $\lambda_j - \lambda_1 \geq 0$. Hessian is PSD $\Longrightarrow$ local (and global) minimum.

-$\lambda = \lambda_n$ (maximum eigenvalue): $A - \lambda_n I$ has eigenvalues $\lambda_j - \lambda_n \leq 0$. Hessian is NSD $\Longrightarrow$ local (and global) maximum.

-$\lambda = \lambda_k$, $1 < k < n$: $A - \lambda_k I$ has both positive and negative eigenvalues $\Longrightarrow$ saddle point.